

LAB 2 WIRESHARK 封包觀察

OUTLINE

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 - Lab 2-1, Lab 2-2
 - Note:Your jobs
 - Please upload your Lab Exercise results to the Tronclass couse website (one file per group)

實驗目的

- 了解網路協定(network protocol)在網路環境中所扮演的角色
- 透過實際觀察封包的組成及特定協定的運作，了解通訊協定層和網路運作機制。
- 熟悉網路封包分析軟體(WireShark)的操作流程。

WIRESHARK 簡介

- Wireshark 是免費開源的網路封包分析軟體，網路封包分析軟體的功能是截取網路封包，並盡可能顯示出最為詳細的網路封包資料。
- 如果電腦沒有WireShark軟體，可至<https://www.wireshark.org/download.html>下載

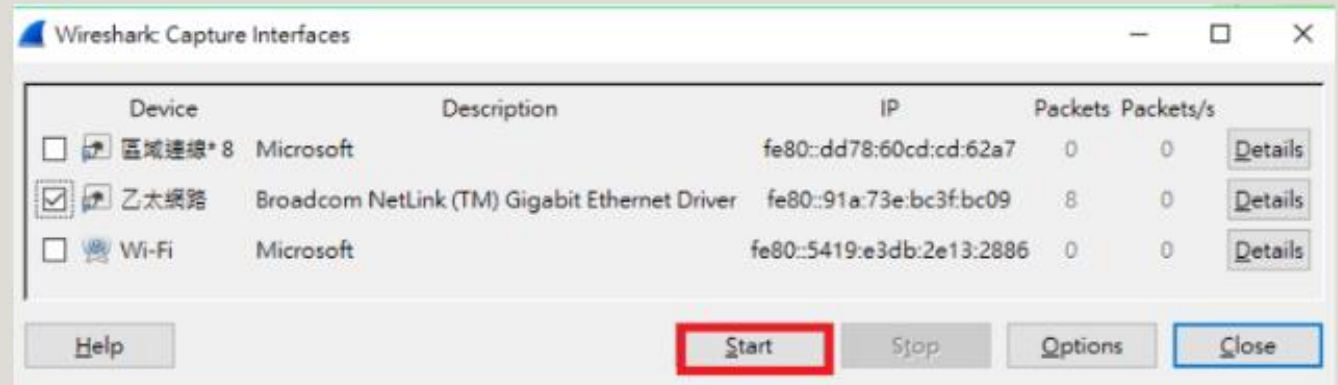
WIRESHARK 介面使用介紹

► 操作

✎ 執行、接收封包資料

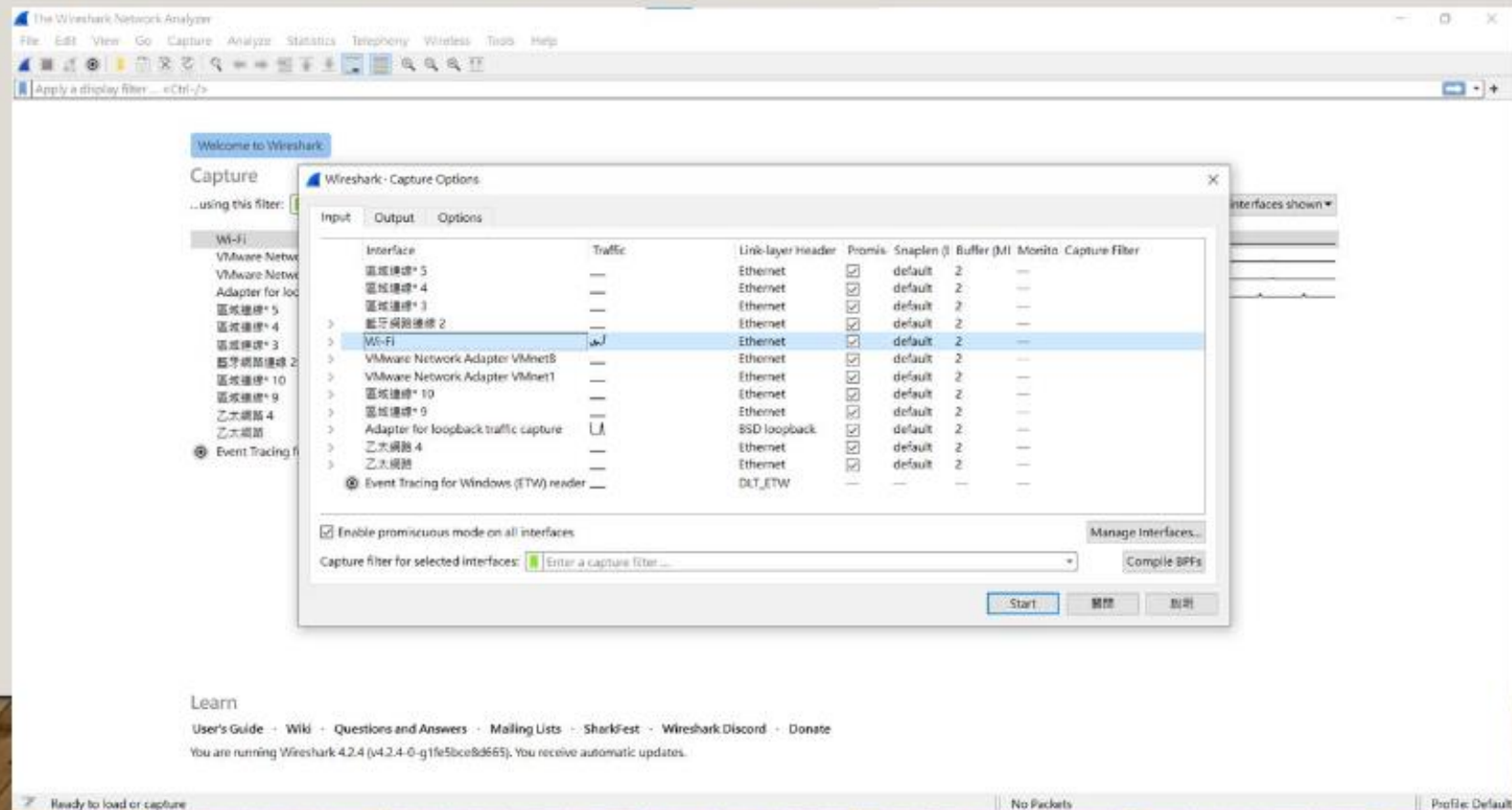


✎ 暫停、看目前封包值



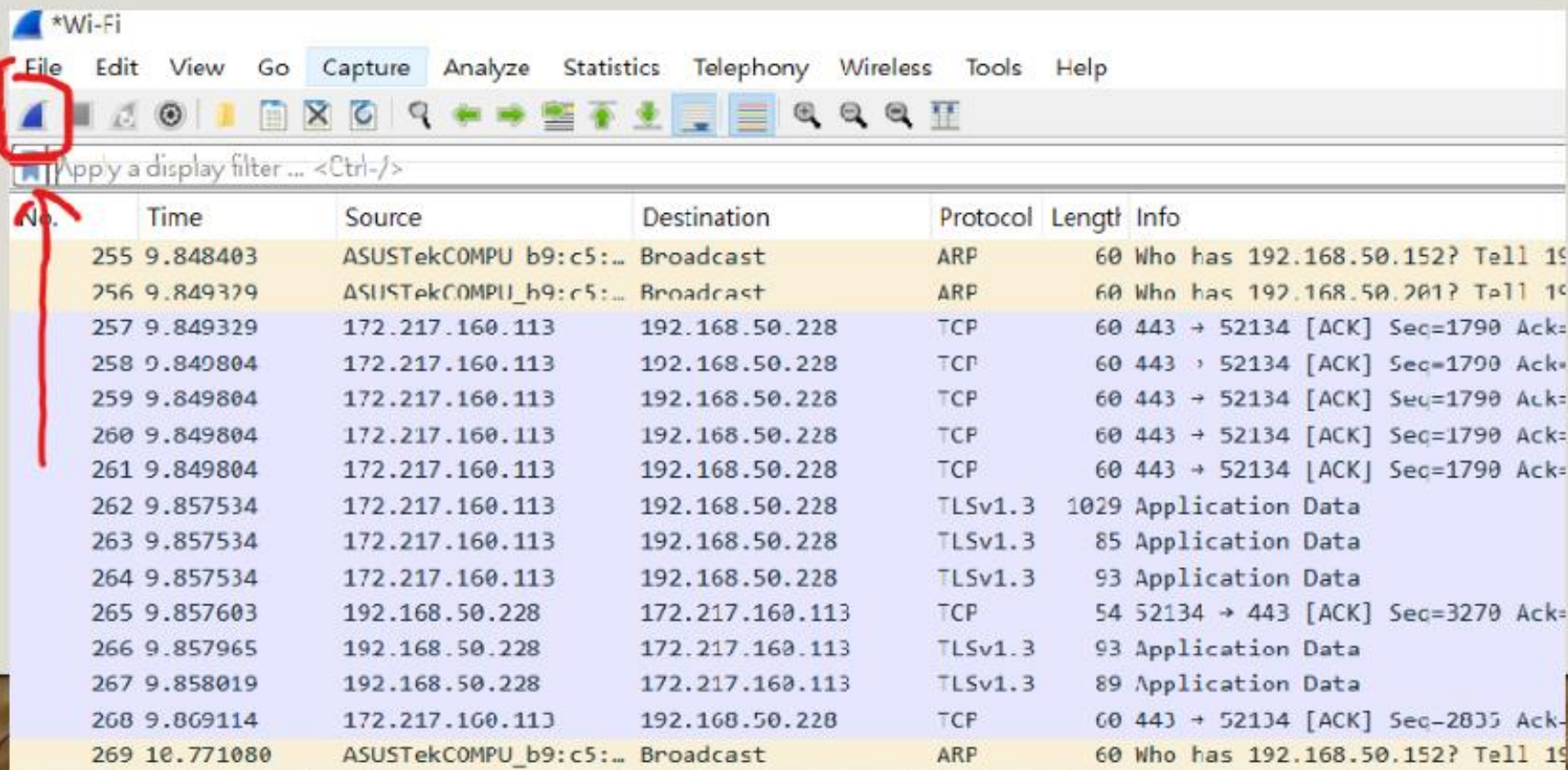
選擇網路

- 點選capture以選取合適的網路 註:範例是用Wi-Fi



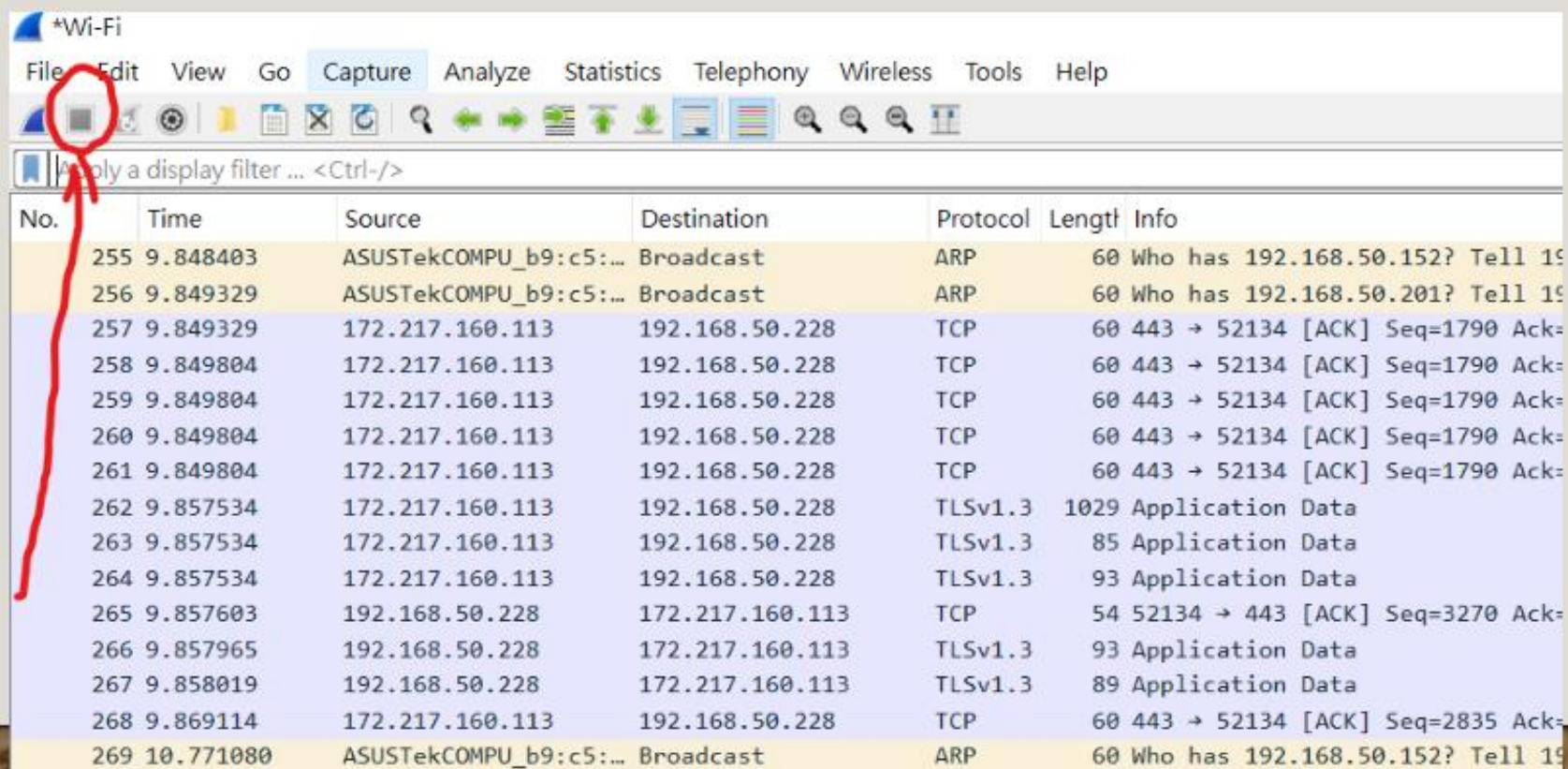
開啟封包接收

- 點選左上的魚鰭狀圖案，開始執行封包接收



暫停封包接收

- 點選左上方正方形圖案暫停封包接收



查看封包内容

- 以下(紅色框內)可以查看封包的詳細內容

The image shows a Wireshark packet capture interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main window is divided into three panes: a packet list on the left, a packet details pane in the middle, and a packet bytes pane on the right.

The packet list pane shows a list of captured packets. The selected packet is 250, which is highlighted in blue. The details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The bytes pane shows the raw data of the packet in hexadecimal and ASCII.

The packet list pane shows the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
255	9.849604	ASUSTekCOMPJ_b9:c5:...	Broadcast	ARP	60	Who has 192.168.50.152? Tell 192.168.50.151
256	9.849604	ASUSTekCOMPJ_b9:c5:...	Broadcast	ARP	60	Who has 192.168.50.201? Tell 192.168.50.151
257	9.849629	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack=2705 Win=74240 Len=0
258	9.849604	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack=2744 Win=74240 Len=0
259	9.849604	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack=2779 Win=74240 Len=0
260	9.849604	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack=2872 Win=74240 Len=0
261	9.849604	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=1790 Ack=3778 Win=77856 Len=0
262	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	1829	Application Data
263	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	85	Application Data
264	9.857534	172.217.160.113	192.168.50.228	TLSv1.3	93	Application Data
265	9.857603	192.168.50.228	172.217.160.113	TCP	54	52134 → 443 [ACK] Seq=3270 Ack=2835 Win=65324 Len=0
266	9.857965	192.168.50.228	172.217.160.113	TLSv1.3	93	Application Data
267	9.858019	192.168.50.228	172.217.160.113	TLSv1.3	89	Application Data
268	9.869114	172.217.160.113	192.168.50.228	TCP	60	443 → 52134 [ACK] Seq=2835 Ack=3344 Win=77856 Len=0
269	10.771080	ASUSTekCOMPJ_b9:c5:...	Broadcast	ARP	60	Who has 192.168.50.152? Tell 192.168.50.151
270	10.771080	ASUSTekCOMPJ_b9:c5:...	Broadcast	ARP	60	Who has 192.168.50.201? Tell 192.168.50.151
271	10.869719	192.168.50.228	192.168.50.228	TCP	60	52134 → 6080 [SYN] Seq=0 Win=0 Len=0 MSS=1460 ssthresh=256 SACK_PERM
272	10.963375	192.159.136.234	192.168.50.228	TLSv1.2	141	Application Data

The details pane for packet 250 shows the following structure:

- Ethernet II, Src: ASUSTekCOMPJ_53:fe:50 (08:bf:b0:53:fe:50), Dst: Intel_ac:1d:31 (d4:d2:52:ac:1d:31)
- Internet Protocol Version 4, Src: 172.217.160.113, Dst: 192.168.50.228
- Transmission Control Protocol, Src Port: 443, Dst Port: 52134, Seq: 1790, Ack: 2872, Len: 0

The bytes pane shows the raw data of the packet in hexadecimal and ASCII:

```
0000  d4 c2 52 ac 1d 31 00 0f  60 53 fe 50 00 00 45 00
0010  00 28 13 40 00 00 37 06  41 b8 ac d9 a0 71 c0 a5
0020  37 e4 01 b0 ch a5 fa 1d  27 c3 26 fa 54 c9 50 10
0030  01 22 08 a5 00 00 00 00  00 00 00 00
```

作業繳交

- Wireshark 頁面必須含有**TCP**封包，並截圖

